
Software Requirements Specification

for

Rail and Road-based Supply Chain Distribution System

Version 1.0 approved

Prepared by Group 24

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Revision History

Name	Date	Reason For Changes	Version
Group 24	25/07/2026	Initial submission covering sections 1-6 and appendices	1.0

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to specify the software and database requirements for Version 1.0 of the **Kandypack Rail and Road-based Supply Chain Distribution System**. This document details the database-driven logistics platform designed to replace legacy Excel-based tracking systems, covering bulk rail transportation, last-mile truck dispatching, roster management, and managerial reporting.

1.2 Document Conventions

Priority Levels: Functional requirements are marked with priorities (**High, Medium, Low**). Higher-level requirements pass their priority to child requirements unless explicitly noted.

Database Objects: Names of database tables, procedures, functions, and triggers (e.g., RosterDetail, Order_Date) are written in monospaced font.

Business Constraints: Key constraints (e.g., mandatory lead times, weekly hour limits) are explicitly highlighted in **bold**.

1.3 Intended Audience and Reading Suggestions

This document is written for:

- **Database Developers & Engineers:** To guide schema design, primary/foreign key definitions, stored procedures, triggers, and indexing strategies.
- **Software Developers:** To understand REST API and JDBC data interaction requirements for backend and web interface integration.
- **Quality Assurance (QA) Testers:** To formulate test cases validating ACID transactions, capacity overflow, and roster constraint rules.
- **Project Managers & Course Evaluators:** To verify system scope, deliverables, and domain modelling alignment.

1.4 Product Scope

The Kandypack Logistics Platform is a database-driven distribution system designed for Kandypack, an FMCG manufacturing company based in Kandy, Sri Lanka. The platform handles bulk rail distribution between Kandy and six major destination cities (namely: Colombo, Negombo, Galle, Matara, Jaffna and Trincomalee) and replaces existing manual Excel tracking to resolve delivery inefficiencies caused by growing order volumes.

1.5 References

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.
- Kandypack Supply Chain & Logistics Operations Specifications (Project 3 Brief).
- Sri Lanka Railways Fixed Cargo & Route Schedule Guidelines.

2. Overall Description

2.1 Product Perspective

The system we have built is a new, self-contained product. Currently, Kandypack uses a manual Excel-based tracking system, which this platform replaces entirely — it is not a follow-on version of any existing software. It is a web-based platform: a relational database (accessed via JDBC) sits behind a Spring Boot service layer, which exposes REST APIs to a web-based interface used by internal staff.

2.2 Product Functions

The functions provided by this system for staff, at a high level, are to:

Customer & Order Management:

- Create and validate customer orders, matching each to the delivery route covering its address
- Register orders containing multiple line items, each with a specific space consumption rate

Rail Cargo Scheduling & Overflow:

- Schedule order items onto each cargo trip, checking the capacity that can be carried per trip and sending any overflow to the next scheduled trip

Store & Last-Mile Delivery Operations:

- Keep records and receipts of stores containing goods
- Assign trucks, drivers, and assistants to scheduled routes

Driver & Assistant Roster Management:

- Enforce proper driver and assistant rostering rules (assigning them with respect to weekly working-hour limits and consecutive-trip constraints)

Delivery Tracking & Reporting:

- Track and update the status of each delivery item
- Generate the management reports under Section 4.5

2.3 User Classes and Characteristics

These are the user classes expected:

- Order Clerks — create and manage customer orders; frequently used, moderate technical expertise
- Logistics / Station Managers — personnel at headquarters and destination station stores overseeing train unloading, truck/route assignment, and delivery status updates; frequently used, central to last-mile operations
- Roster / Fleet Coordinators — staff managing driver and assistant work assignments, creating duty rosters and assigning them to truck schedules; frequently used
- Executive Management — company executives requiring high-level operational and financial insight; read-only access to reporting, used infrequently
- Database Administrator (DBA) — maintains schema, indexes, stored procedures, and backups; used infrequently, high technical expertise

2.4 Operating Environment

This system requires MySQL 8.0+ as the database server, running on Linux or Windows Server. The application server runs on JRE 17+, hosting a Spring Boot REST API application. Clients access the system through a modern web browser (Google Chrome, Firefox, or Microsoft Edge) over standard HTTP/HTTPS. Headquarters and destination

store hubs are connected over a LAN/WAN using TCP/IP. No offline or mobile operation is supported for Version 1.0.

2.5 Design and Implementation Constraints

- The system must use MySQL 8.0+ and a Spring Boot/JRE 17+ backend, as fixed implementation choices for this version
- ACID properties must be present at the database level, using stored procedures, functions, and triggers where appropriate (checking capacity requirements and roster conflicts)
- Primary and foreign keys must be defined on all entities to maintain proper references to and from them
- Indexes must be applied on columns used for heavy filtering and reporting (order date, destination city, or route identifier)
- The system must contain test data with at least 40 orders, 10 distinct routes, and valid train schedules with predefined capacities sufficient to exercise all test scenarios

2.6 User Documentation

A data dictionary (Appendix A: Glossary) is provided with this SRS. As the current scope of the UI is basic, a separate user manual is not planned for Version 1.0; on-screen labels and tooltips are considered sufficient (refer to Appendix C).

2.7 Assumptions and Dependencies

- Each defined product type has a fixed space consumption rate (cargo units per item), backed by product data
- Each trip has a fixed total cargo capacity according to the Sri Lanka Railways schedule
- Each delivery route is predefined and does not overlap with other delivery routes; one truck is scheduled per route, and each route has a defined maximum delivery time
- For weekly driver/assistant time calculations, each delivery duration is derived directly from the defined maximum delivery time of its assigned route
- There is one receiving store for each destination city unless specified otherwise
- Currency for reporting is Sri Lankan Rupees (LKR); quantities are tracked in boxes or units, consistent with each product's defined space consumption rate

3. External Interface Requirements

3.1 User Interfaces

The Kandypack integrated Logistics Platform provides a web-based graphical user interface (GUI) that enables different users to perform their tasks efficiently. Each user is provided with role-based access to the system after authentication.

The interface includes the following modules:

- **Login Screen**
 - Secure authentication using username and password.
 - Redirects users according to their assigned role.
- **Dashboard**
 - Provides a summary of current system activities, including pending orders, upcoming train trips, truck schedules, inventory levels, and recent deliveries.
- **Customer Interface**
 - Allows customers to register and log in to the system.
 - Allows customers to manage their profile information.
 - Enables customers to view their order history and delivery status.
- **Order Management Interface**
 - Allows customers to place new orders by selecting products and specifying quantities.
 - Supports orders containing multiple products with different quantities.
 - Validates that orders are placed at least seven days before the requested delivery date.
- **Train Schedule Interface**
 - Displays available train trips, destination cities, allocated cargo space, remaining capacity, and overflow schedules.
- **Store & Inventory Interface**
 - Shows products available at each destination warehouse.
 - Updates inventory automatically after train arrival and truck dispatch.
- **Truck Dispatch Interface**
 - Allows logistics coordinators to assign trucks, drivers, assistants, and delivery routes.
 - Displays route duration and delivery status.

- **Reports Interface**

- Enables management to generate quarterly sales reports, truck utilization reports, employee working hours, customer order history, and city-wise sales analysis.

The interface should remain responsive on modern desktop web browsers and provide clear error messages whenever invalid data is entered.

3.2 Hardware Interfaces

The system interacts with standard computer hardware available in the Kandypack head office and regional stores.

The hardware interfaces include:

Hardware	Purpose
Desktop/Laptop Computers	Accessing the web application
Central Database Server	Stores all operational data
Network Router	Communication between headquarters and regional stores

The application communicates with these devices through standard operating system drivers and TCP/IP networking.

3.3 Software Interfaces

The Kandypack Logistics Platform integrates with several software components to perform its operations.

Software Component	Interface	Purpose
MySQL Database Server	JDBC	Stores all logistics data and executes SQL queries
Backend Application	REST API	Processes business logic and communicates with the database
Web Browser	HTTP/HTTPS	Provides access to the user interface

Operating System (Windows/Linux)	System Services	File management, networking, and security
Reporting Module	SQL Queries / Report Viewer	Generates management reports and displays them to users

The backend application communicates with the relational database to store and retrieve logistics data, while the frontend exchanges data with the backend through standard web requests.

3.4 Communications Interfaces

The system communicates over a secure local or internet network using standard communication protocols.

The communication interfaces include:

- HTTP and HTTPS protocols for communication between the web browser and the application server.
- TCP/IP networking for communication between the headquarters and destination stores.
- Secure database connections between the backend server and the MySQL database.

4. System Features

4.1 Train Cargo Capacity & Space Consumption Scheduling

4.1.1 Description and Priority

- Description : Determines the utilization of train capacities based on the product specific space consumption and schedules bulk shipments using the train routes of Sri Lanka Railways. Manages Overflow: Schedules overbooked items for their next trip on the train.
- Priority : High

4.1.2 Stimulus/Response Sequences

- System/User : Allocates a train for pending bulk orders for Kandy origin.
- System : Can sum up all the space necessary using rates of space per item (such as 0.5 space per box).

- System : Allocates a fixed cargo capacity for the target train trip
- System : Allocates capacity until it is maximum and overflows quantity to the second available train schedule.

4.1.3 Functional Requirements

- REQ-1 : The system shall use the following formula to determine the amount of space being used: $\text{Total Space} = \text{Sum of Quantity}_i * \text{Space Consumption Rate}_i$.
- REQ-2 : Overbooking of trains shall not be allowed in the system via database transactions or procedures.
- REQ-3 : When the volume of orders exceeds the available cargo space, the system should divide the over-cargo order and/or overload orders in order to be sent to the next scheduled trip

4.2 Order Placement & Route Matching

4.2.1 Description and Priority

- Description : Enables customers/users to order multiple items with at least 7 days advance notice. The system automatically matches the delivery address with a predefined delivery route.
- Priority : High

4.2.2 Stimulus/Response Sequences

- User : Selects delivery address, items and the delivery date (No earlier than 7 days in advance).
- System : Supports lead time (A minimum of 7 days).
- System : Creates candidates for destination delivery based on the predefined route maps.
- User : To approve and place orders.
- System : Saves the order in the system and places it in the status Placed.

4.2.3 Functional Requirements

- REQ-1 : Place an order at least one week in advance.

- REQ-2 : A system should be able to handle multiple orders with references to product items and quantity.
- REQ-3 : Orders should automatically be assigned to a pre-defined route based on the destination delivery added.

4.3 Last-Mile Delivery & Truck Scheduling

4.3.1 Description and Priority

- Description : Operates dispatch services for trucks taking goods from their warehouse destinations at a rail route location to their final destinations.
- Priority : High

4.3.2 Stimulus/Response Sequences

- User : Picks orders out of queue and schedules them to a destination store that are not loaded.
- System : Displays and shows trucks available, assigned routes, maximum Delivery Time and constraints.
- User : Schedules deliveries with a truck and route.
- System : Provides updates of shipment status and keeps a record of the expected delivery completion times.

4.3.3 Functional Requirements

- REQ-1 : The system needs the ability to arrange the dispatches of trucks only from stores where the destination train stations are located.
- REQ-2 : Each truck schedule should be within the maximum allowed delivery times for the trucks on that schedule.
- REQ-3 : The system shall be capable of changing the order status, e.g. In-Transit and Delivered once deliveries have taken place.

4.4 Driver & Assistant Roster & Constraint Management

4.4.1 Description and Priority

- Description : Assigns drivers and assistants to truck schedules while strictly validating working-hour caps and consecutive assignment constraints
- Priority : High

4.4.2 Stimulus/Response Sequences

- User : Schedules a driver & driver assistant for a truck delivery.
- System : Ensures that weekly working hours are within permissible limits for drivers (40 hrs) & assistant (60 hrs).
- System : Ensures two successive shift rules (Driver- no two successive shifts, Assistant- max 2 successive shifts).
- System : Confirms assignment if valid, or blocks assignment with an error prompt if constraints are violated.

4.4.3 Functional Requirements

- REQ-1 : Every truck schedule needs to have only one driver assigned, and only one driver assistant.
- REQ-2 : The system should not allow a driver to be scheduled to make two back-to-back truck trips.
- REQ-3 : The system should not allow an assistant to be scheduled to deliver more than two routes in a row.
- REQ-4 : The system shall enforce a maximum of 40 scheduled hours per week for drivers.
- REQ-5 : The system shall enforce a maximum of 60 scheduled hours per week for assistants.
- REQ-6 : No two time slots can be assigned to the same truck, driver or assistant.

4.5 Management Reporting & Analytics

4.5.1 Description and Priority

- Description : Prepares operating, financial and utilization reports for management decision making.
- Priority : Medium

4.5.2 Stimulus/Response Sequences

- Manager : Switches to the reports module and clicks on the report type and parameters (e.g. Quarter, Month and City).
- System : The main purpose of this system is to perform backend stored procedures/queries.
- System : Outputs formatted tabular and/or graphic reports to the screen.

4.5.3 Functional Requirements

- REQ-1 : Quarterly Sales Report: Generate Sales Breakdown by Value-Volume for each Quarter
- REQ-2 : Top Items: Report and list those most ordered products for a given quarter.
- REQ-3 : Geographic Sales: Provide City-wise / Route-wise sales performance break down.
- REQ-4 : Working Hours: will create weekly/monthly working hours summaries for drivers & assistants.
- REQ-5 : Fleet Usage: Produce truck utilization analysis for a month.
- REQ-6 : Customer History: Output detailed order history of customers to include details of fulfillment & delivery.

4.6 Database Integrity & ACID Enforcement (System Internal Feature)

4.6.1 Description and Priority

- Description : System-level capability that enforces consistency, key relationships and triggers and constraint checks in the database.
- Priority : High

4.6.2 Stimulus/Response Sequences

- System/User Action : User executes database transactions such as placing orders, assigning deliveries and updating schedules.

- Database Engine : Runs constraints, foreign key checks, triggers, and transactions.
- System : Commits changes when ACID conditions are met, or rolls back changes on exception.

4.6.3 Functional Requirements

- REQ-1 : The system shall adopt explicit ACID compliant transactions for the scheduling of train capacity and for booking delivery slots.
- REQ-2 : Triggers run automatically to check rosters for constraints before new rosters are inserted into RosterDetail.
- REQ-3 : For reporting queries, indexes should usually be added on frequently queried fields (e.g. Route_ID, Order_Date or Driver_ID) to improve the performance of such queries.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- **Transactional Response Time**

All standard daily operations, like entering new orders, checking delivery routes, and updating order status, must finish within 1.5 seconds during regular working hours.

- **Data Capacity and Scalability**

The database should be designed and indexed properly so that it continues to run fast even after storing more than 100,000 order records and delivery details.

- **Report Generation Speed**

Complex reports, such as quarterly sales summaries and monthly truck usage analysis, must be generated within 3 to 5 seconds even when processing a full year of sales data.

- **System Concurrency**

The system must handle 30 to 50 users working at the same time, like store managers and dispatch staff, without slowing down or locking up the database.

5.2 Safety Requirements

- **Database Transaction Safety**

The system must follow strict ACID properties for all operations, such as assigning orders to trains or trucks, so that data is never lost or partially saved if there is a sudden power loss or system failure.

- **Operational and Workload Safety**

The database must automatically check and enforce maximum working hour rules for drivers and assistants to prevent fatigue and ensure safety during road deliveries.

- **Backup and Data Recovery**

The system should create automated daily backups of the database to make sure no operational data or customer order history is lost in case of unexpected hardware damage.

5.3 Security Requirements

- **User Authentication and Role-Based Access**

Users must log in with valid credentials. The system must restrict access based on user roles so that normal staff cannot modify master data or override roster limits.

- **Session Security and Timeout**

The system must automatically log out inactive users after 15 minutes of idle time to prevent unauthorized access from unattended computers in offices or delivery hubs

- **Sensitive Data Encryption**

Passwords and sensitive customer information stored in the database must be encrypted using secure hashing algorithms to protect data from unauthorized internal or external access.

- **Database Action Auditing**

Important system actions, such as changing delivery statuses, modifying driver assignments, or updating order details, must be recorded in an audit log with the user ID and timestamp.

5.4 Software Quality Attributes

- **System Reliability**

The database must maintain continuous availability and strict transaction integrity so that order scheduling and cargo allocation never fail during peak business hours.

- **Database Maintainability**

The relational structure must follow proper normalization standards (3NF) so that new tables, delivery routes, or features can be added in the future without breaking existing queries.

- **Operational Usability**

The system interface should show clear error messages and guidance when a user violates business rules, such as trying to schedule an order with less than 7 days lead time.

- **Data Portability and Compatibility**

The database schema must be designed using standard SQL so that it can be backed up, restored, or migrated across different relational database management systems without data loss.

5.5 Business Rules

- **Advance Order Placement Rule**

Customers must place all orders at least 7 days before the expected delivery date so that train capacity and truck delivery routes can be planned properly.

- **Driver Workload and Roster Rule**

A driver cannot be scheduled for two consecutive truck delivery trips, and their total working time must not exceed 40 hours in a single week.

- **Assistant Workload and Roster Rule**

A driver assistant can be assigned to a maximum of two consecutive delivery trips, and their total working time must not exceed 60 hours in a single week.

- **Train Capacity and Cargo Overflow Rule**

Fixed cargo capacity allocated by Sri Lanka Railways must be respected based on product space consumption rates. If an order exceeds the available train space, the excess quantity must automatically roll over to the next available trip.

- **Schedule Conflict Prevention Rule**

The system must prevent assigning the same truck, driver, or assistant to overlapping route times or conflicting delivery schedules.

6. Other Requirements

- **Relational Integrity and Normalization**

The database schema must be organized using proper Primary Keys and Foreign Keys to maintain data consistency and prevent orphaned records across tables.

- **Database Stored Logic (Procedures, Functions, and Triggers)**

Automated database triggers and procedures must be implemented to enforce business rules, such as checking driver weekly hours, preventing double-booking, and calculating train capacity before saving a transaction.

- **Indexing Strategy**

B-tree indexes must be applied on frequently queried columns, such as order date, route ID, delivery status, driver ID, and train trip ID, to speed up search and report operations.

- **Test Data Provisioning**

The database must be populated with initial test data including at least 40 customer orders, 10 delivery routes, and complete train schedules with defined capacities for testing.

Appendix A: Glossary

- **FMCG (Fast-Moving Consumer Goods)**

High-turnover consumer products that require fast, reliable transportation and scheduling across rail and road distribution channels.

- **Lead Time**

The mandatory 7-day advance booking window required between when an order is placed and when it is scheduled for delivery.

- **Rollover Cargo**

The portion of order volume that exceeds a train trip's allocated capacity limit and is automatically shifted to the next available train schedule.

- **Consecutive Trips**

Back-to-back delivery assignments for drivers or assistants without an intervening rest period or off-duty interval.

Appendix B: Analysis Models

- **Entity Relationship Diagram (ERD)**

Illustrates the normalized relational database design, showing primary key to foreign key mappings between orders, customers, train schedules, truck routes, and employee rosters.

- **Data Flow Diagram (DFD)**

Shows the flow of information from customer order entry through train capacity allocation, rollover handling, truck routing, and driver roster validation.

- **Process Flowchart**

Visualizes the step-by-step logic for validating driver weekly hour limits, checking consecutive trip rules, and assigning overflow cargo to subsequent train schedules.

Appendix C: Data Dictionary / Entity Tables

Customer

A person or business that places orders with Kandypack.

Attribute	Type	Key	Description
CustomerID	INT	PK	Unique identifier for the customer
FullName	VARCHAR(100)		Customer's full name / business name
Address	VARCHAR(200)		Street address
City	VARCHAR(50)		City of the customer
ContactNumber	VARCHAR(15)		Phone number
Email	VARCHAR(100)		Email address
username	VARCHAR(50)		Username used to log in to the system
password	VARCHAR(255)		Encrypted password for customer authentication

Order

A customer order for one or more products.

Attribute	Type	Key	Description
OrderID	INT	PK	Unique identifier for the order
CustomerID	INT	FK	References Customer.CustomerID
OrderDate	DATE		Date the order was placed
Status	VARCHAR(20)		Pending / Shipped / Delivered / Cancelled
TotalAmount	DECIMAL(10, 2)		Total order value
RouteID	INT	FK	References Route.RouteID
StoreID	INT	FK	References Store.StoreID

OrderDetail

A single product line within an order (order-product junction).

Attribute	Type	Key	Description
OrderDetailID	INT	PK	Unique identifier for the order line
OrderID	INT	FK	References Order.OrderID
ProductID	INT	FK	References Product.ProductID
Quantity	INT		Units of the product ordered
LineTotal	DECIMAL(10, 2)		Total amount for the order line, calculated as $\text{Quantity} \times \text{UnitPrice}$.

Product

An FMCG product that Kandypack distributes.

Attribute	Type	Key	Description
ProductID	INT	PK	Unique identifier for the product
ProductName	VARCHAR(100)		Name of the product
Category	VARCHAR(50)		Product category
UnitPrice	DECIMAL(10, 2)		Standard selling price
SpaceConsumption	DECIMAL(6, 2)		Volume/space used per unit (for capacity checks)
StockQuantity	INT		Number of product units currently available in stock.

Store

A regional distribution hub (e.g. Colombo, Negombo, Galle, Matara, Jaffna, Trincomalee) where goods arrive by rail before last-mile trucking.

Attribute	Type	Key	Description
StoreID	INT	PK	Unique identifier for the store
StoreName	VARCHAR(100)		Name of the regional warehouse/store
City	VARCHAR(50)		City where the store is located
Capacity	INT		Maximum storage capacity

Route

A rail route connecting the main Kandy warehouse to a regional store.

Attribute	Type	Key	Description
RouteID	INT	PK	Unique identifier for the route
StoreID	INT	FK	References Store.StoreID
Distance	DECIMAL(8, 2)		Distance in kilometres

Train

A train is used to move goods along rail routes.

Attribute	Type	Key	Description
TrainID	INT	PK	Unique identifier for the train
TrainName	VARCHAR(50)		Name/number of the train
Capacity	INT		Maximum cargo capacity

TrainSchedule

A scheduled run of a specific train along a specific route.

Attribute	Type	Key	Description
ScheduleID	INT	PK	Unique identifier for the schedule entry
TrainID	INT	FK	References Train.TrainID
RouteID	INT	FK	References Route.RouteID
DepartureTime	TIME		Scheduled departure time

Attribute	Type	Key	Description
ArrivalTime	TIME		Scheduled arrival time
DayOfWeek	VARCHAR(10)		Day the schedule runs

Shipment

The portion of an order's goods carried on a particular train schedule.

Attribute	Type	Key	Description
ShipmentID	INT	PK	Unique identifier for the shipment
OrderID	INT	FK	References Order.OrderID
ScheduleID	INT	FK	References TrainSchedule.ScheduleID
Quantity	INT		Units of the order carried on this train run
ShipmentDate	DATE		The date on which the shipment is transported from the origin to the destination

Truck

A last-mile delivery truck based at a regional store.

Attribute	Type	Key	Description
TruckID	INT	PK	Unique identifier for the truck
RegistrationNumber	VARCHAR(20)		Vehicle registration number
Capacity	INT		Maximum cargo capacity
StoreID	INT	FK	References Store.StoreID (home store)

Driver

A driver who operates delivery trucks.

Attribute	Type	Key	Description
DriverID	INT	PK	Unique identifier for the driver
Name	VARCHAR(100)		Driver's full name
LicenseNumber	VARCHAR(20)		Driving license number
ContactNumber	VARCHAR(15)		Phone number

Assistant

An assistant who accompanies a driver on deliveries.

Attribute	Type	Key	Description
AssistantID	INT	PK	Unique identifier for the assistant
Name	VARCHAR(100)		Assistant's full name
ContactNumber	VARCHAR(15)		Phone number

Truck_Trip

A scheduled truck dispatch: one truck, one route, one driver, and one assistant, for a given time window. Deliveries reference this trip rather than the truck/driver/assistant directly.

Attribute	Type	Key	Description
TripID	INT	PK	Unique identifier for the truck trip
TruckID	INT	FK	References Truck.TruckID
RouteID	INT	FK	References Route.RouteID
DriverID	INT	FK	References Driver.DriverID
AssistantID	INT	FK	References Assistant.AssistantID
DispatchTime	TIME		Scheduled dispatch time
ReturnTime	TIME		Scheduled/expected return time
TripDate	DATE		The scheduled date on which the truck trip takes place

Delivery

The last-mile delivery of an order from a store to the customer.

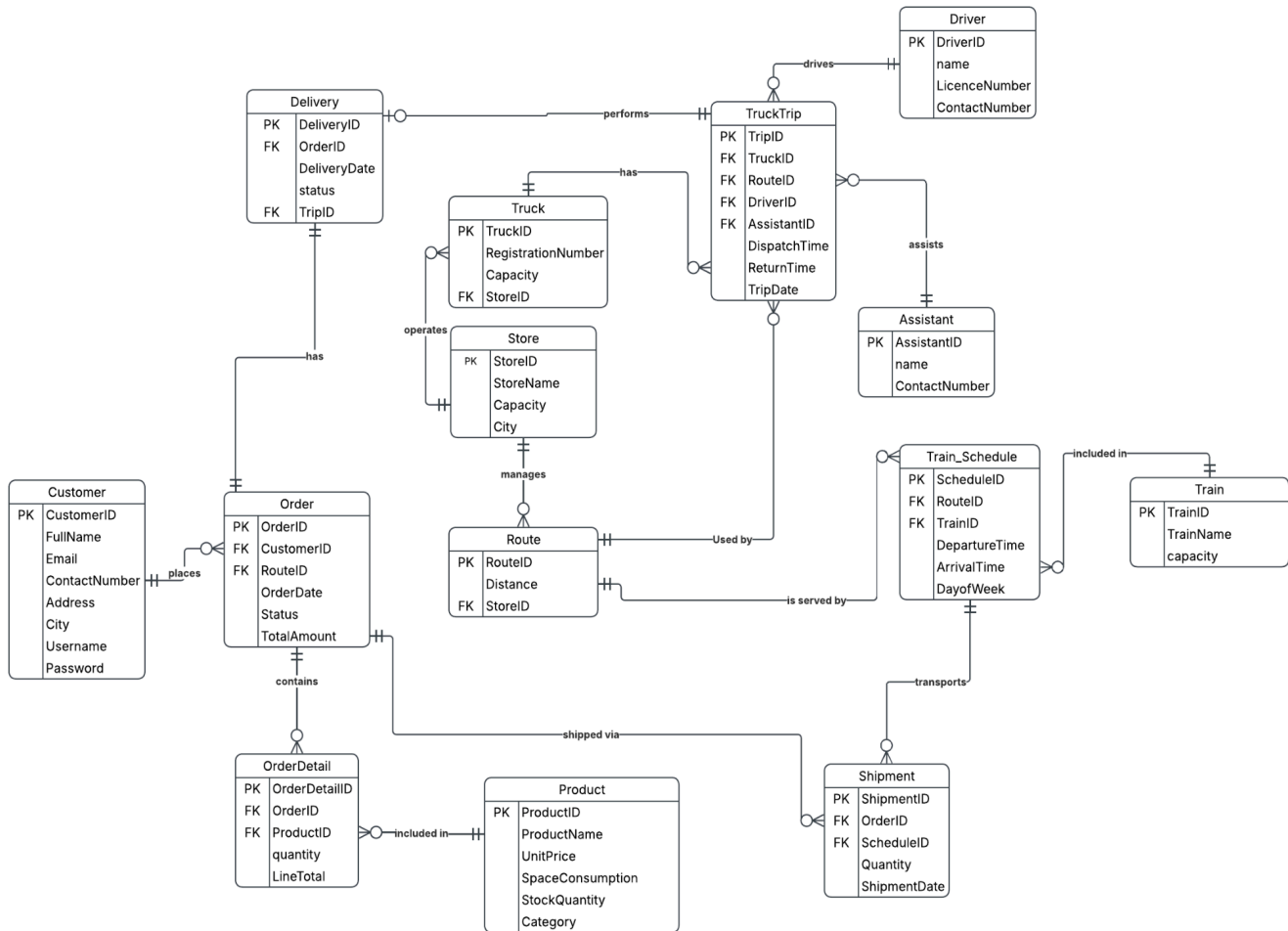
Attribute	Type	Key	Description
DeliveryID	INT	PK	Unique identifier for the delivery
OrderID	INT	FK	References Order.OrderID
TripID	INT	FK	References Truck_Trip.TripID
DeliveryDate	DATE		Date of delivery
Status	VARCHAR(20)		Scheduled / In Transit / Delivered

Entity Relationships

The table below summarises the relationships between entities and their cardinality.

Entity A	Entity B	Cardinality	Description
Customer	Order	1 : M	A customer places many orders
Order	OrderDetail	1 : M	An order contains many order lines
Product	OrderDetail	1 : M	A product appears on many order lines
Store	Route	1 : M	A store is the destination of many routes (Kandy is the fixed origin for all routes)
Route	TrainSchedule	1 : M	A route has many scheduled train runs
Train	TrainSchedule	1 : M	A train has many scheduled runs
Order	Shipment	1 : M	An order may be split across shipments
TrainSchedule	Shipment	1 : M	A scheduled run carries many shipments
Store	Truck	1 : M	A store has many trucks based there
Order	Delivery	1 : 1	An order results in one last-mile delivery
Truck_Trip	Delivery	1 : M	A truck trip may serve multiple deliveries
Truck	Truck_Trip	1 : M	A truck performs many trips
Driver	Truck_Trip	1 : M	A driver drives many trips
Assistant	Truck_Trip	1 : M	An assistant supports many trips
Route	Truck_Trip	1 : M	A route has many scheduled truck trips

ER Diagram



Appendix D: To Be Determined List

- **Rail Capacity Threshold Specifications**

Exact volume and weight limits per train compartment provided by Sri Lanka Railways need formal confirmation to set the precise baseline for cargo overflow triggers.

- **Emergency Overtime Overrides**

The specific administrative process and authorization role required to override the 40-hour driver limit during severe operational delays or emergencies need final approval.

- **System Database Host Selection**

The exact database server hardware, cloud provider, and operating system specifications for hosting the production database remain to be finalized by the infrastructure team.

- **External ERP Integration Protocol**

The exact API format and data transmission frequency for syncing order data with existing legacy accounting tools need to be defined

- **Driver / Assistant System Access**

Whether drivers and assistants require individual system logins, or remain data-only subjects tracked for rostering purposes, needs confirmation (see section 2.3 and Section 2.7).